Reported injury rates depend on exposure

Independent research program 31 October 2026 UTC

Not peer reviewed. Current submitted summary data; descriptive sensitivity.

Practical finding

A pooled injury rate can be dominated by a few reported exposure values. In this fixed manufacturing cohort, the five highest-hour records account for 92.53% of eligible reported hours, while carrying 1 reported case. This concentration audit is explicitly retrospective: it was added after the frozen comparison exposed the denominator imbalance.

The full eligible cohort reports 115,164 cases over 361,383,028,589 hours, a pooled rate of 0.063735 per 100 full-time-equivalent worker-years. These are source-accounting values, not a validated population rate or evidence that the exposure values are correct. Both independent source parsers agree with the captured native values; that agreement does not establish source truth.

Recommended use

Audit exposure concentration before using a pooled reported rate to center smoothing. A dashboard should retain the numerator, reported hours, missingness and smoothing assumptions. Seek source clarification for extreme exposure values before operational use. Do not silently cap or remove observations and call the result corrected.

The original source and all frozen estimates remain unchanged. Conditional rates after removing the highest-hour records are presented as accounting diagnostics, not cleaned estimates or a recommendation to delete those records. No employer identities or rankings are public.

Source and fixed scope

The official OSHA 2025 Summary Data file contains 400,288 rows. The fixed cohort retains 46,530 manufacturing records in broad NAICS prefixes 31, 32 and 33 with reported maximum-size codes for 20–99, 100–249 or unsplit 20–249 employees. Annual average employment is not the size selector. 46,511 records have a consistent known numerator and positive reported hours; 19 nonpositive-hour records remain explicitly outside the rate denominator.

The numerical design was committed after the file was downloaded but before establishment values were parsed. Only the header, size and hash had been inspected. Values were visible during source reconciliation; this was source informed, not blinded. One current posted file does not establish original submission timing, physical-establishment identity or prediction-time availability.

Complete fixed accounts and conditional diagnostics

NAICS	Maximum size	Eligible	Excluded	Zero cases	Cases	Raw rate
31	20-99	3,571	0	1,257	6,662	0.258
31	100-249	2,203	1	194	13,386	0.268
31	20-249 unsplit	476	0	135	1,211	2.908
32	20-99	9,637	11	3,754	15,339	2.423
32	100-249	4,073	2	650	17,060	2.109
32	20-249 unsplit	2,961	1	1,161	5,536	0.003
33	20-99	13,654	2	5,515	22,368	1.296
33	100-249	6,122	2	1,000	26,366	2.160
33	20-249 unsplit	3,814	0	1,551	7,236	1.958

Rates are reported cases per 100 full-time-equivalent worker-years, using 200,000 hours. Every fixed stratum is retained, including the unsplit legacy size class. NAICS versions can differ. Source-account categories, exact exposures and rates remain in the complete exports; no annual-average substitution or missing-to-zero fill is used.

All declared descriptive shrinkage strengths

Let $t = h/200{,}000$ and $\mu = \sum k / \sum t$ across the complete eligible cohort. A Gamma–Poisson working model uses Gamma shape $b\mu$ and rate b, giving mean $(k + b\mu)/(t + b)$. The fixed prior exposures $b \in \{0.1, 1, 10\}$ span 20,000–2,000,000 hours. They are authored sensitivity probes, not tuned or empirically validated defaults. The center is estimated from this same cohort.

The authored boundary is rate $> 2\mu$, with no regulatory or safety meaning. The raw rate exceeds it for 31,183 records. At $b = 0.1, 1, 10$, respectively, 0, 2 and 55 move to at/below the boundary; none moves upward. All 27 stratum/strength accounts and 135,204 sufficient-cell/strength cases are exported. No winner is selected and no future accuracy is measured.

Retrospective concentration audit: all four cases

Top records	Top cases	Hours share	Remaining hours	Conditional rate
1	1	38.78%	221,241,028,589	0.104
5	1	92.53%	26,984,308,589	0.854
10	10	95.86%	14,978,817,537	1.538
100	599	97.83%	7,843,757,569	2.921

All four conditional rates use the remaining submitted records. They are retrospective diagnostics, not replacement population estimates. The largest single reported exposure is 140,142,000,000 hours. No source error is asserted without clarification.

For zero recorded cases, $-\log(0.05)/t$ is the one-sided 95% upper rate under a homogeneous independent Poisson process. All 15,217 zero-case records and complete stratum bounds are retained. This assumption-based illustration has no measured interval calibration or workplace-safety validity.

Verification, implementation and limits

Independent evidence

Separate pandas C parsing, SQLite key multiplicities and exact Fraction tokens reconcile all native rows in 445,371 finite conditions. Saved native-bound source checks pass 315,540. Independent weighted SQLite and exact-rational reconstruction checks every account, cell/strength, strict boundary, switch direction and request in 1,004,148 conditions. The retrospective record-expanded SQLite concentration audit passes 40 conditions.

The original committed freeze bound 21 actual dependencies before comparisons. A disclosed post-analysis gzip-container amendment changes one writer, preserving every numeric result and the original code/freeze in HISTORY. Byte-preserving Git checkout fixes an inherited Windows line-ending failure. Both isolated full-history CPython 3.10.11 and 3.12.14 replays reproduce six files exactly, with all six source-directory files unchanged. Failed compatibility evidence is retained.

All 31 authored fixture methods pass, covering maximum versus average employment, documented injury-flag codes, duplicate groups, missing/invalid counts, zero/nonpositive hours, exact rate ties, prior strengths and unsupported inferences. The inspected figures bind 62 actual SVG bars and 248 coordinate conditions. Finite checks are not universal correctness guarantees.

Runnable implementation exercise

Seven requests return two qualified reported-stratum accounts and reject safety ranking, legal fault, worker eligibility, future risk and denominator substitution. The interface retains numerator, exposure and explicit unknown safety/future accuracy. It does not certify arbitrary caller data or a production service.

```
python -m pip install -r requirements.txt
python src/replay.py --out ../injury-rate-replay
python src/check_diagnostics.py
```

Use a new output directory. The replay uses saved counts; it does not redownload mutable native data.

Material limits and next evidence

The original year-held-out partial-pooling proposal remains unexecuted. This is a same-cohort-centered descriptive sensitivity study, with no protected test, future-risk accuracy or new estimator. Source reporting, optional/nonrequired submissions, mixed industry-code versions and unknown physical-establishment identity limit generalization. Reporting does not establish fault, violation or benefit eligibility. No causal, production-use or realized-savings claim is made.

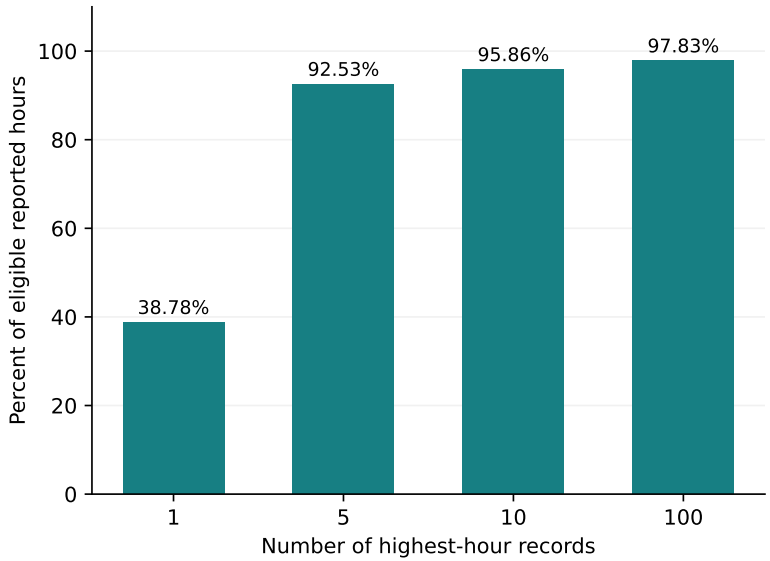
Raw establishment names, addresses, EINs, identifiers and native files remain private. No case-detail or worker data were requested. Separate formal-publication projects, including Azure optimization, remain private. Seek corrected or clarified exposure and verified identity/dated availability before a predictive extension.

Source: OSHA Injury Tracking Application, 2025 Summary Data, submitted January–August 2026, captured 31 October 2026 UTC. https://www.osha.gov/itadata. Definitions: April 2024 Summary Data Dictionary and January 2025 ITA Data Users Guide. Exact URL, native hash, source accounts and earlier access failures are retained. OSHA does not endorse this independent analysis. MIT covers original code; government facts are attributed separately.

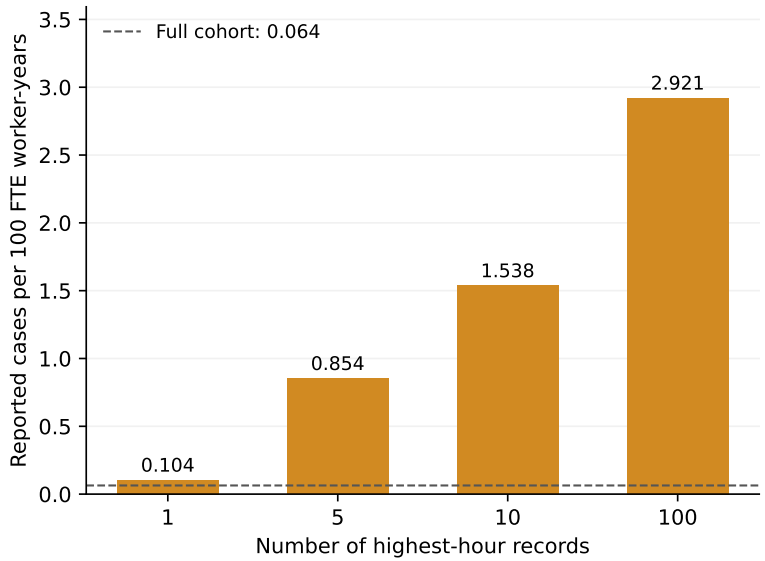
Five of 46,511 records carry most reported hours

Retrospective exposure audit: all four top-k cases retained; full source and frozen estimates unchanged

Share held by the top records



Rate among the remaining records

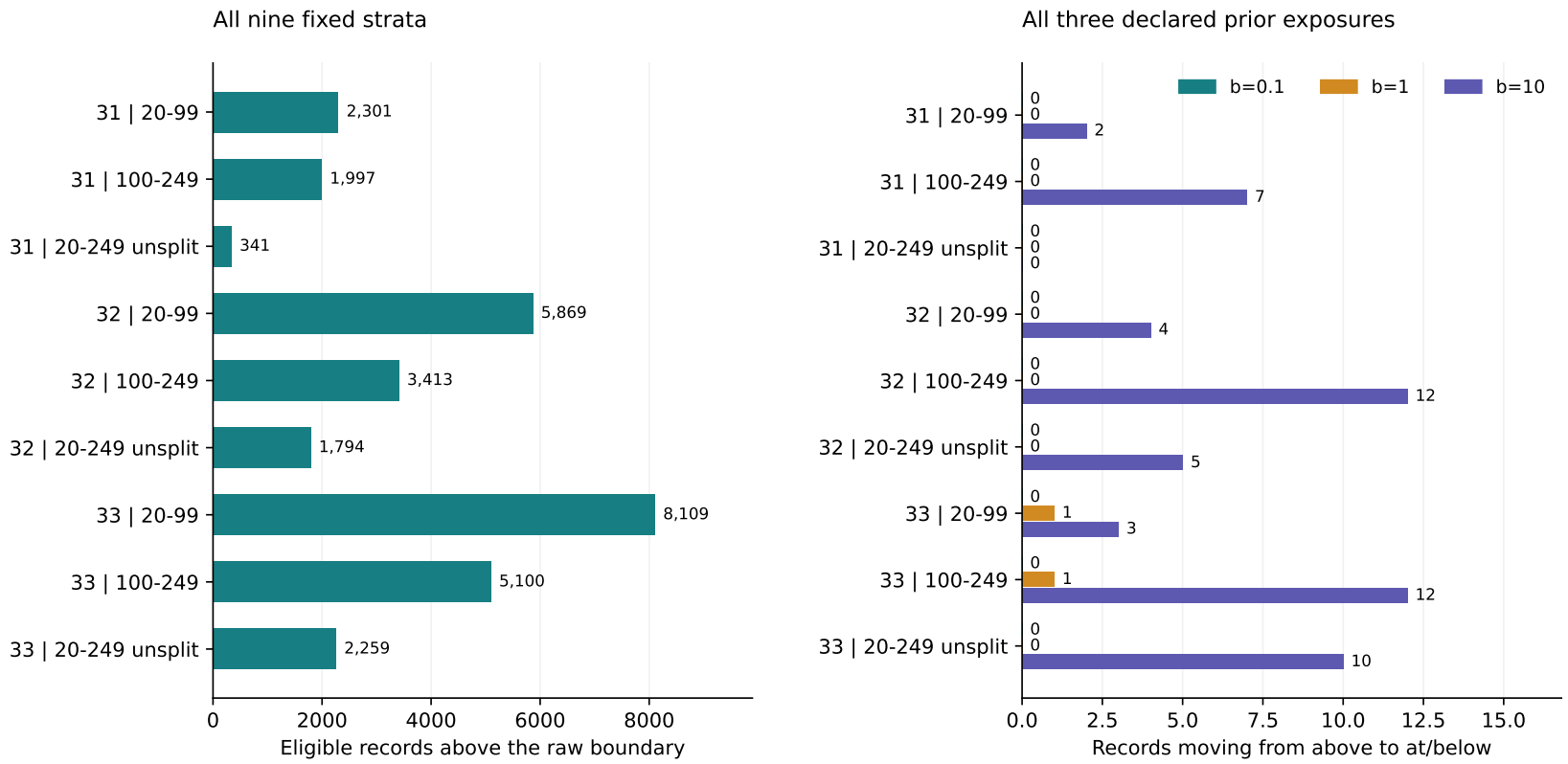


Conditional rates are accounting diagnostics, not corrected data or replacement population estimates.

Complete declared smoothing comparisons

Declared smoothing assumptions change a reported-rate label

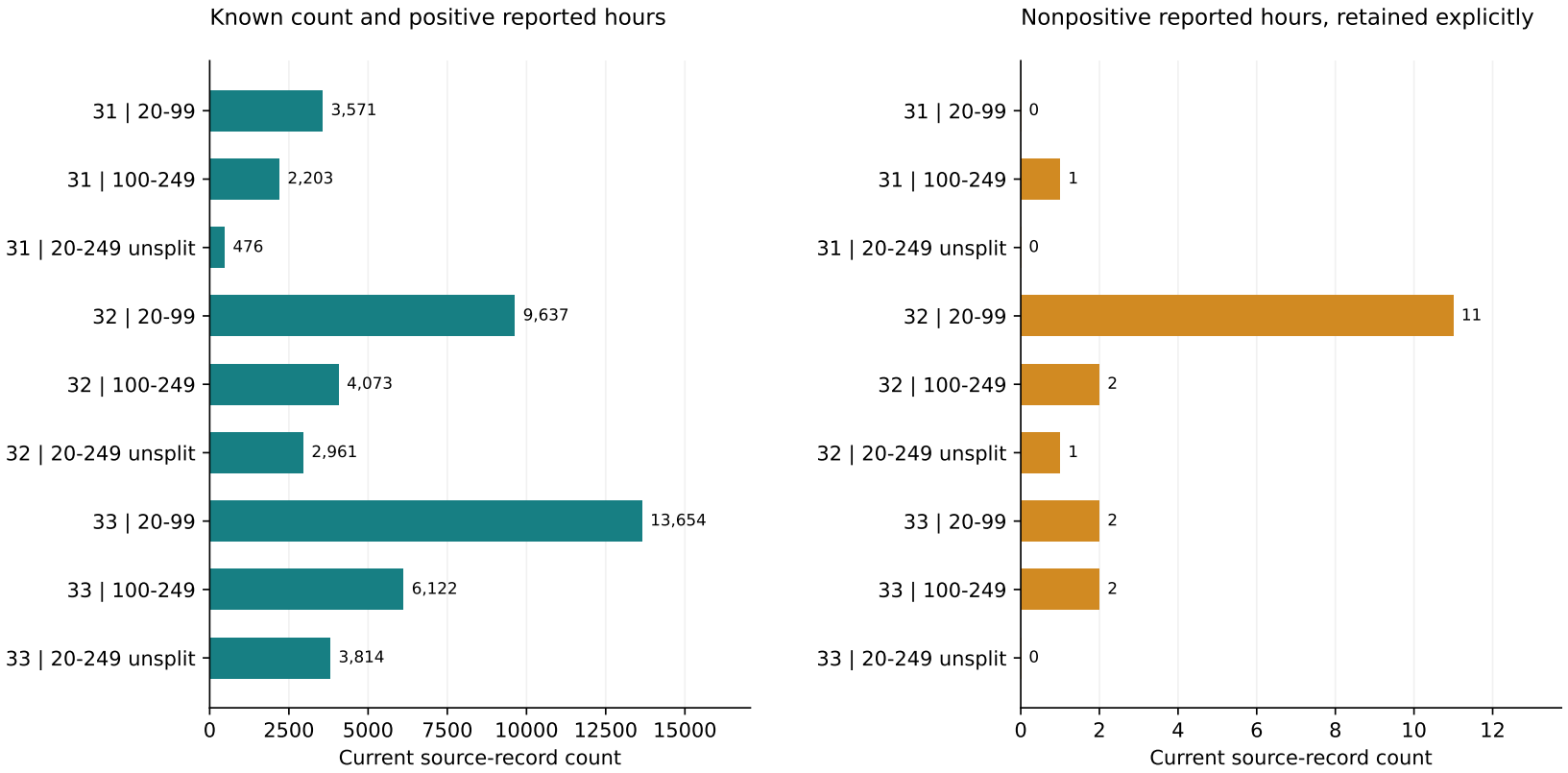
Authored boundary: rate > twice the same-cohort pooled rate. No safety meaning or selected prior strength.



Size uses reported maximum employment. Broad NAICS prefixes mix source code versions; no employer ranks.

Nonpositive hours remain outside the rate denominator

Complete fixed manufacturing cohort: 46,530 source records, 46,511 eligible and 19 with nonpositive hours



Source: OSHA ITA 2025 Summary Data. No unknown value is filled with zero; source accuracy is not established.